ESE 2150 Lab 3

Experimental Verification of Nodal and Mesh Analysis

Gonzalo Cuervo Ritvik Ellendula

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A. Nodal Analysis: Experimental Verification

A.1. Measurement Procedure

The circuit was constructed on a breadboard using 100 Ω, 470 Ω, and 1 kΩ resistors. Power was supplied using a Keysight E36441A DC Power Supply with a set current limit of 0.1 A to protect against short circuits. Node voltages V_1 and V_2 were measured relative to ground using a digital multimeter. Branch currents I_x, I_y, and I_z were measured in series using the ammeter.

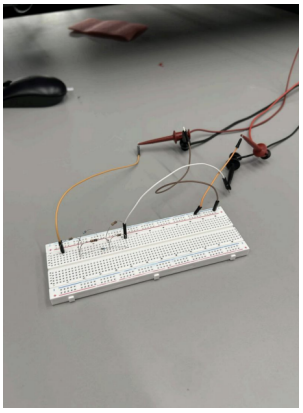


Figure 1: Breadboard setup for Part A.

A.2. Measurement and Calculation Results

Quantity	V_1 (V)	V_2 (V)	I_x (mA)	I_y (mA)	I_z (mA)
Measurement	1.12	−0.735	1.8	−0.684	2.3
Nodal analysis	1.123	−0.740	1.863	−0.740	2.389
Percentage (%)	−0.27	−0.68	−3.50	−8.19	−3.87

Table 1: Reported measurement and nodal-analysis results.

The lab handout defines percentage error as

$$\text{Percentage error} = \left| \frac{\text{Measured result} - \text{Calculated result}}{\text{Measured result}} \right| \times 100\%.$$

A.3. Nodal Analysis Calculations

Applying Kirchhoff's current law at V_1 gives

$$\begin{aligned}\frac{V_1-2}{100}+\frac{V_1-V_2}{1000}+\frac{V_1}{470}+\frac{V_1+1}{470}&=0,\\ \frac{11V_1-V_2-20}{1000}+\frac{2V_1+1}{470}&=0.\end{aligned}$$

Applying Kirchhoff's current law at V_2 gives

$$\begin{aligned}\frac{V_2}{1000}+\frac{V_2-V_1}{1000}+\frac{V_2+1}{100}&=0,\\ V_2+V_2-V_1+10V_2+10&=0,\\ 12V_2-V_1+10&=0.\end{aligned}$$

Solving the two node equations gives the reported results

$$\boxed{V_1=1.123\,\mathrm{V}},\qquad \boxed{V_2=-0.740\,\mathrm{V}}.$$

The branch currents are

$$\begin{aligned}I_x&=\frac{V_1-V_2}{1000}=\frac{1.123+0.740}{1000}=1.863\,\mathrm{mA},\\ I_y&=\frac{V_2}{1000}=\frac{-0.740}{1000}=-0.740\,\mathrm{mA},\\ I_z&=\frac{V_1}{470}=\frac{1.123}{470}=2.389\,\mathrm{mA}.\end{aligned}$$

A.4. Discussion

Minor differences exist between measured and calculated values. This is due to standard resistor tolerances, internal non-zero resistance of the ammeter during current measurement, and small contact resistances across breadboard connections since they are all real-world components and cannot behave ideally.

B. Mesh Analysis: Experimental Verification

B.1. Measurement Procedure

The circuit shown in Figure 4 of the lab handout was assembled using 100 Ω, 470 Ω, 680 Ω, and 1 kΩ resistors. Voltages at nodes V_1, V_2, and V_3 were measured with a digital multimeter referenced to the ground node. Branch currents I_w, I_x, I_y, and I_z were measured with the ammeter placed in series with each respective branch.

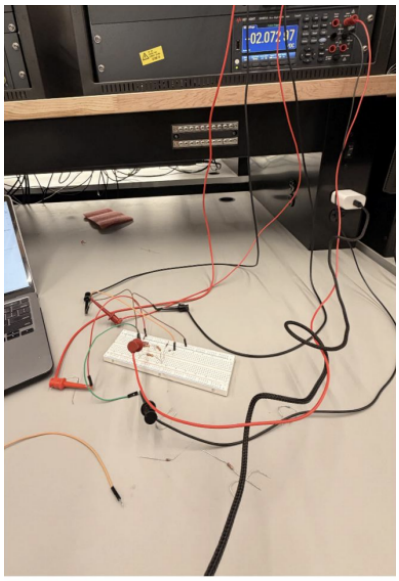


Figure 2: Breadboard setup and measurement equipment for Part B.

B.2. Measurement and Calculation Results

Quantity	V_1 (V)	V_2 (V)	V_3 (V)	I_w (mA)	I_x (mA)	I_y (mA)	I_z (mA)
Measurement	0.868	0.073	2.306	2.306	0.868	−2.115	−6.94
Mesh analysis	0.863	0.064	2.250	2.250	0.863	−2.039	−7.503
Percentage (%)	0.58	14.06	2.49	2.49	0.58	3.73	7.50

Table 2: Reported measurement and mesh-analysis results.

B.3. Mesh Analysis Calculations

Using the mesh-current directions drawn in the submitted calculations, Kirchhoff's voltage law gives

$$\begin{aligned} 1000I_1 + 680(I_1 - I_4) + 470(I_1 - I_2) - 2 &= 0, \\ 1000(I_2 - I_3) + 2 + 470(I_2 - I_1) + 680(I_2 - I_4) &= 0, \\ 100I_3 + 3 + 1000(I_3 - I_2) &= 0, \\ 680I_4 + 680(I_4 - I_2) + 680(I_4 - I_1) &= 0. \end{aligned}$$

Here, resistances are in ohms and mesh currents are in amperes. In matrix form,

$$\begin{bmatrix} 2150 & -470 & 0 & -680 \\ -470 & 2150 & -1000 & -680 \\ 0 & -1000 & 1100 & 0 \\ -680 & -680 & 0 & 2040 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -3 \\ 0 \end{bmatrix}.$$

Solving gives the reported mesh currents

$$\begin{aligned} I_1 &= -0.863\,\text{mA}, & I_2 &= -5.254\,\text{mA}, \\ I_3 &= -7.503\,\text{mA}, & I_4 &= -2.039\,\text{mA}. \end{aligned}$$

The branch currents are related to the mesh currents by

$$\begin{aligned} I_w &= I_2 - I_3 = 2.250\,\text{mA}, \\ I_x &= -I_1 = 0.863\,\text{mA}, \\ I_y &= I_4 = -2.039\,\text{mA}, \\ I_z &= I_3 = -7.503\,\text{mA}. \end{aligned}$$

The node voltages follow from

$$\begin{aligned} \frac{V_1}{1000} &= I_x & \Longrightarrow \quad V_1 &= 0.863\,\text{V}, \\ \frac{V_2 + 2}{470} &= I_1 - I_2 & \Longrightarrow \quad V_2 &= 0.064\,\text{V}, \\ \frac{V_3 - 3}{100} &= I_z & \Longrightarrow \quad V_3 &= 2.250\,\text{V}. \end{aligned}$$

B.4. Discussion

Minor differences exist due to resistor tolerances, internal non-zero resistance of the ammeter during current measurement, and small contact resistances across breadboard connections since they are all real-world components and cannot behave ideally.

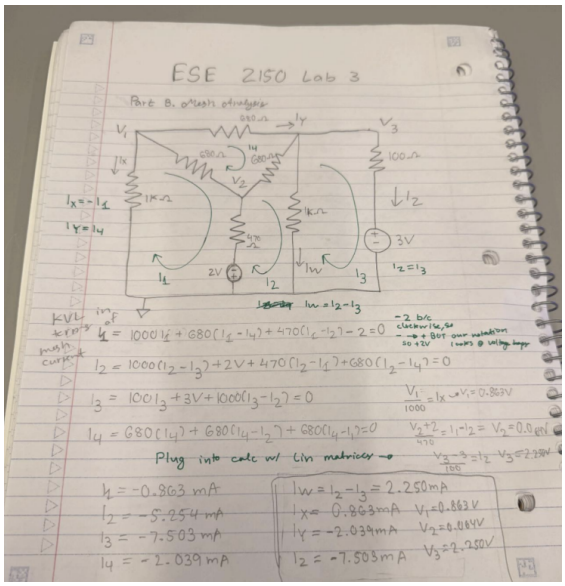


Figure 3: Wheatstone Bridge Mesh Analysis